Boss Challenge — Paper 27 (Class 7)

Motion & Time | Forests | Algebraic Expressions | Comparing Quantities

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

---

1. The distance covered by a moving body in one unit of time is what we call its:

- **(A)** acceleration
- **(B)** speed
- **(C)** oscillation
- **(D)** displacement

2. In a forest, the green plants and trees that make their own food using sunlight are called the:

- **(A)** consumers
- **(B)** decomposers
- **(C)** producers
- **(D)** scavengers

3. A letter such as x or y that can stand for different number values is called a:

- **(A)** coefficient
- **(B)** constant
- **(C)** variable
- **(D)** term

4. The comparison of two quantities by division, written in the form 2 : 3, is called their:

- **(A)** product
- **(B)** average
- **(C)** percentage
- **(D)** ratio

5. When speed is measured using the SI units of distance and time, the unit that comes out is the:

- **(A)** second per metre
- **(B)** kilometre per hour
- **(C)** metre per minute
- **(D)** metre per second (m/s)

6. The microorganisms and fungi that break down dead leaves and dead animals into humus are the forest's:

- **(A)** predators
- **(B)** decomposers
- **(C)** herbivores
- **(D)** producers

7. In the expression $4x + 7$, the number 7, whose value never changes, is called a:

- **(A)** coefficient
- **(B)** factor
- **(C)** constant
- **(D)** variable

8. The symbol % used in percentages is a short way of writing 'out of':

(A) the total
(B) 100
(C) 1000
(D) ten

9. A car that covers exactly 15 metres in every single second, never speeding up or slowing down, is moving with:

(A) no motion
(B) uniform motion
(C) circular motion
(D) non-uniform motion

10. The roof-like covering formed by the branchy tops of the tall forest trees is called the:

(A) understorey
(B) herb layer
(C) canopy
(D) humus

11. The parts of an expression that are added or subtracted, such as 3x and 5 in 3x + 5, are called its:

(A) constants
(B) coefficients
(C) factors
(D) terms

12. The fraction 1/2 written as a percentage is:

(A) 2%
(B) 25%
(C) 100%
(D) 50%

13. A bus that covers different distances in equal time intervals as it weaves through traffic is in:

(A) periodic motion
(B) non-uniform motion
(C) rest
(D) uniform motion

14. The branchy top part of a single tall tree, made of its spreading branches and leaves, is known as its:

(A) root
(B) bark
(C) canopy
(D) crown

15. In the term 6y, the number 6 that is multiplied with the variable y is called the:

(A) constant
(B) exponent
(C) coefficient
(D) variable

16. The percentage 25% written as a fraction in its lowest terms is:

(A) 1/4
(B) 1/2
(C) 4/1
(D) 25/10

17. On a distance-time graph, the motion of an object moving at a steady speed appears as a:

(A) straight slanting line
(B) flat horizontal line
(C) curved line
(D) vertical line

18. The shorter trees and shrubs that grow in the shade below the tall canopy form the layer called the:

(A) canopy
(B) understorey
(C) crown
(D) forest floor

19. The pair 4x and 9x, which have exactly the same variable part, are called:

(A) constants
(B) unlike terms
(C) like terms
(D) factors

20. 20% of 200 is:

(A) 40
(B) 20
(C) 100
(D) 400

21. On a distance-time graph, a flat horizontal line tells us that during that time the object was:

(A) speeding up
(B) moving at steady speed
(C) at rest (not moving)
(D) moving fast

22. The dark, crumbly, nutrient-rich material formed when dead plants and animals rot in the forest is called:

- **(A)** gravel
- **(B)** humus
- **(C)** sand
- **(D)** clay

23. The pair 5x and 5y, which have different variable parts, are called:

- **(A)** equal terms
- **(B)** constants
- **(C)** like terms
- **(D)** unlike terms

24. The decimal 0.75 written as a percentage is:

- **(A)** 7.5%
- **(B)** 750%
- **(C)** 0.75%
- **(D)** 75%

25. The meter on a vehicle's dashboard that shows how fast it is going at that moment is the:

- **(A)** odometer
- **(B)** barometer
- **(C)** speedometer
- **(D)** thermometer

26. Forests are called the green lungs of the Earth because they absorb carbon dioxide and release:

- **(A)** smoke
- **(B)** nitrogen
- **(C)** carbon dioxide
- **(D)** oxygen

27. An algebraic expression that has only one term, such as 7xy, is called a:

- **(A)** binomial
- **(B)** polynomial of four terms
- **(C)** trinomial
- **(D)** monomial

28. Which of these is an equivalent ratio to 2 : 3?

- **(A)** 4 : 6
- **(B)** 2 : 6
- **(C)** 4 : 3
- **(D)** 3 : 2

29. The meter on a vehicle that keeps adding up the total distance the vehicle has travelled is the:

- **(A)** speedometer
- **(B)** compass
- **(C)** stopwatch
- **(D)** odometer

30. Tree roots grip the soil firmly and stop it from being washed away by rain; forests therefore prevent:

- **(A)** evaporation
- **(B)** photosynthesis
- **(C)** germination
- **(D)** soil erosion

31. An algebraic expression with exactly two terms, such as $3x + 5$, is called a:

- **(A)** binomial
- **(B)** constant
- **(C)** monomial
- **(D)** trinomial

32. The ratio 10 : 15 written in its simplest form is:

- **(A)** 5 : 3
- **(B)** 2 : 3
- **(C)** 3 : 2
- **(D)** 10 : 15

33. One complete to-and-fro swing of a simple pendulum, from one extreme back to the same extreme, is called one:

- **(A)** oscillation
- **(B)** rotation
- **(C)** revolution
- **(D)** vibration of sound

34. A chain that shows who eats whom, such as grass -> deer -> tiger, is called a:

- **(A)** water cycle
- **(B)** food web
- **(C)** life cycle
- **(D)** food chain

35. An algebraic expression with exactly three terms, such as $x + y + 7$, is called a:

- **(A)** trinomial
- **(B)** binomial
- **(C)** monomial
- **(D)** variable

36. A cyclist's speed rises from 10 km/h to 12 km/h. The percentage increase in her speed is:

(A) 20%
(B) 2%
(C) 120%
(D) 12%

37. The time a simple pendulum takes to finish one complete oscillation is known as its:

(A) frequency
(B) time period
(C) amplitude
(D) speed

38. In the chain grass -> deer -> tiger, the deer that eats only plants is a:

(A) decomposer
(B) herbivore
(C) carnivore
(D) producer

39. In the term 5xy, the quantities 5, x and y that are multiplied together to make it are its:

(A) constants
(B) exponents
(C) factors
(D) terms

40. A shopkeeper buys a bag for ■200 and sells it for ■250. His profit percent is:

(A) 25%
(B) 12.5%
(C) 20%
(D) 50%

41. The basic SI unit used to measure time is the:

(A) day
(B) hour
(C) minute
(D) second

42. In the chain grass -> deer -> tiger, the tiger that hunts and eats other animals is a:

(A) herbivore
(B) carnivore
(C) producer
(D) scavenger

43. Adding the like terms 3a + 5a gives:

(A) 8a
(B) 15a
(C) 8
(D) $8a^2$

44. An item bought for ■500 is sold for ■400. The loss percent is:

- **(A)** 20%
- **(B)** 25%
- **(C)** 100%
- **(D)** 10%

45. A leopard runs at 36 km/h. Expressed in metres per second (divide by 3.6), this speed is:

- **(A)** 36 m/s
- **(B)** 10 m/s
- **(C)** 100 m/s
- **(D)** 3.6 m/s

46. The cutting down of forest trees on a large scale, clearing the land of its trees, is called:

- **(A)** afforestation
- **(B)** deforestation
- **(C)** pollination
- **(D)** germination

47. Subtracting 4m from 9m gives:

- **(A)** 13m
- **(B)** $5m^2$
- **(C)** 5m
- **(D)** 5

48. The extra money paid for the use of borrowed money over a fixed time is called:

- **(A)** discount
- **(B)** simple interest
- **(C)** principal
- **(D)** profit

49. A forest ranger's jeep covers 60 km along a fire-line in 2 hours. Its speed is:

- **(A)** 30 km/h
- **(B)** 60 km/h
- **(C)** 120 km/h
- **(D)** 15 km/h

50. Forests help bring rain because the water vapour given out by their trees adds moisture to the:

- **(A)** food chain
- **(B)** rock cycle
- **(C)** water cycle
- **(D)** life cycle

51. The algebraic expression for 'a number x increased by 5' is:

- **(A)** x + 5
- **(B)** 5x
- **(C)** x - 5
- **(D)** x / 5

52. In the simple-interest formula, the product of principal, rate and time is divided by:

(A) 100
(B) 1000
(C) 10
(D) the time

53. A deer dashes at 12 m/s for 5 seconds. The distance it covers is:

(A) 12 m
(B) 17 m
(C) 2.4 m
(D) 60 m

54. Decomposers are vital in a forest because they return the nutrients locked in dead matter back to the:

(A) Sun
(B) clouds
(C) bark of trees
(D) soil

55. The algebraic expression for 'three less than twice a number n' is:

(A) 3n - 2
(B) 2n + 3
(C) 2n - 3
(D) 3 - 2n

56. The simple interest on ■1000 at 5% per year for 2 years is:

(A) ■100
(B) ■1000
(C) ■50
(D) ■200

57. A boy cycles 1200 m to school at a steady 4 m/s. The time he takes is:

(A) 4800 s
(B) 1200 s
(C) 300 s
(D) 30 s

58. Because its trees regrow and the forest can renew itself if cared for, a forest is described as a:

(A) man-made resource
(B) renewable resource
(C) non-renewable resource
(D) fossil resource

59. A jeep's speed is (2t + 5) km/h after t hours of driving. At t = 3, its speed is:

- **(A)** 16 km/h
- **(B)** 25 km/h
- **(C)** 10 km/h
- **(D)** 11 km/h

60. If 5 identical pens cost ■50, then by the unitary method 1 pen costs:

- **(A)** ■5
- **(B)** ■250
- **(C)** ■10
- **(D)** ■50

61. Object P covers 100 m in 10 s while object Q covers the same 100 m in 20 s. The faster object is:

- **(A)** cannot be decided
- **(B)** both equal
- **(C)** Q
- **(D)** P

62. A forest gives many animals their food, shelter and space to live; for them the forest is a:

- **(A)** predator
- **(B)** habitat
- **(C)** producer
- **(D)** food chain

63. The value of the expression 3x + 4 when x = 2 is:

- **(A)** 18
- **(B)** 9
- **(C)** 10
- **(D)** 14

64. Converting the percentage 65% into a decimal number gives:

- **(A)** 0.065
- **(B)** 65
- **(C)** 0.65
- **(D)** 6.5

65. A trekker walks 6 km in the first hour and 4 km in the next hour through a forest. His average speed is:

- **(A)** 2 km/h
- **(B)** 6 km/h
- **(C)** 5 km/h
- **(D)** 10 km/h

66. By soaking up rainwater and slowing the run-off down slopes, forests help to prevent:

(A) floods
(B) photosynthesis
(C) erosion only
(D) sunrise

67. A forest patch holds (8n + 3) saplings, where n is the number of rows. The coefficient of n is:

(A) 3
(B) 11
(C) 8
(D) n

68. If 3 out of every 4 children in a class passed a test, the percentage that passed is:

(A) 34%
(B) 75%
(C) 25%
(D) 43%

69. A pendulum completes 20 oscillations in 40 seconds. The time for one oscillation is:

(A) 2 s
(B) 40 s
(C) 0.5 s
(D) 20 s

70. Forests keep the air balanced by taking in carbon dioxide and giving out oxygen during the process of:

(A) digestion
(B) respiration
(C) photosynthesis
(D) evaporation

71. Simplifying 2x + 3x + 4 by collecting like terms gives:

(A) 5x + 4x
(B) 5x + 4
(C) 9x
(D) 6x

72. In a forest plot of 200 trees, 30% are sal trees. The number of sal trees is:

(A) 600
(B) 30
(C) 170
(D) 60

73. On the same distance-time graph two objects are drawn. The one whose line is steeper is moving:

- **(A)** faster
- **(B)** backwards
- **(C)** slower
- **(D)** at rest

74. Honey, gum, wood and medicinal plants that people gather from a forest are all examples of forest:

- **(A)** products
- **(B)** predators
- **(C)** decomposers
- **(D)** habitats

75. An equilateral triangle has each side equal to s. The expression for its perimeter is:

- **(A)** $s + 3$
- **(B)** $3s$
- **(C)** $3 + s$
- **(D)** s^3

76. A toy's price falls from ■50 to ■40. The percentage decrease in its price is:

- **(A)** 25%
- **(B)** 20%
- **(C)** 40%
- **(D)** 10%

77. A pendulum clock keeps good time because a pendulum's swing repeats after equal intervals; such repeating motion is called:

- **(A)** random motion
- **(B)** uniform straight motion
- **(C)** rest
- **(D)** periodic motion

78. The constant falling and rotting of leaves keeps the forest floor covered with a dark layer that is rich in:

- **(A)** nutrients
- **(B)** salt
- **(C)** plastic
- **(D)** metal

79. From the list 7x, 3y, 2x, 5y, the like terms 7x and 2x can be added to give:

- **(A)** 9xy
- **(B)** 9x
- **(C)** 14x
- **(D)** 9

80. If 10% of a number is 8, then the whole number is:

- **(A)** 80
- **(B)** 800
- **(C)** 18
- **(D)** 8

81. Which is the greater speed: 1 m/s or 1 km/h? (Remember 1 m/s = 3.6 km/h.)

- **(A)** cannot compare
- **(B)** 1 km/h
- **(C)** 1 m/s
- **(D)** they are equal

82. In most forest food chains there are far more plants than top carnivores, so the number of producers is the:

- **(A)** equal to carnivores
- **(B)** greatest
- **(C)** smallest
- **(D)** zero

83. A pen costs ■p. The expression for the cost of 6 such pens is:

- **(A)** 6p
- **(B)** p / 6
- **(C)** p■
- **(D)** p + 6

84. The fraction 3/5 expressed as a percentage is:

- **(A)** 30%
- **(B)** 35%
- **(C)** 53%
- **(D)** 60%

85. Average speed for a whole journey is found by dividing the total distance covered by the:

- **(A)** total time taken
- **(B)** starting speed
- **(C)** highest speed reached
- **(D)** number of stops

86. Animals such as vultures that feed on the bodies of already-dead animals are called:

- **(A)** scavengers
- **(B)** decomposers
- **(C)** herbivores
- **(D)** producers

87. How many terms are there in the expression $4x^2 + 3x + 9$?

(A) one
(B) four
(C) three
(D) two

88. Trees and shrubs in a forest patch are in the ratio 3 : 2. If there are 30 in all, the number of trees is:

(A) 10
(B) 12
(C) 15
(D) 18

89. In a simple pendulum, the small heavy object tied to the end of the thread is called the:

(A) axis
(B) bob
(C) dial
(D) pivot

90. Very little sunlight reaches the dark forest floor because most of it is caught and blocked by the:

(A) humus
(B) soil
(C) roots
(D) canopy

91. In the expression $7y - 2$, the constant term is:

(A) y
(B) 7
(C) -2
(D) 2

92. A toy's selling price is ■120 and its cost price is ■100. The profit made is:

(A) ■20
(B) ■100
(C) ■120
(D) ■220

93. A train moves at 72 km/h. Written in metres per second (divide by 3.6), this is:

(A) 36 m/s
(B) 20 m/s
(C) 7.2 m/s
(D) 72 m/s

94. Where forests are removed, less water vapour rises into the air, which over time can reduce the:

(A) gravity
(B) sunlight
(C) rainfall
(D) number of rocks

95. The value of the expression 2a + b when a = 3 and b = 4 is:

(A) 14
(B) 24
(C) 9
(D) 10

96. Out of a 50 km forest trail, a jeep has covered 35 km. The percentage of the trail covered is:

(A) 50%
(B) 65%
(C) 70%
(D) 35%

97. A snail creeps 2 cm in 4 seconds. Its speed in centimetres per second is:

(A) 2 cm/s
(B) 8 cm/s
(C) 4 cm/s
(D) 0.5 cm/s

98. Plants make food, animals eat plants, both die and decompose, returning nutrients to the soil - this repeating flow is the forest's nutrient:

(A) web
(B) cycle
(C) chain
(D) line

99. A forest guard plants t saplings each day. The expression for the number he plants in 7 days is:

(A) t + 7
(B) t■
(C) 7t
(D) t / 7

100. If 4 metres of rope cost ■80, then by the unitary method 7 metres cost:

(A) ■560
(B) ■87
(C) ■20
(D) ■140